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IMAGE PROCESSING APPARATUS CAPABLE OF OBTAINING RESULT OF PROOFREADING OF ORIGINAL WITH SIMPLE OPERATION, CONTROL METHOD, AND STORAGE MEDIUM

Abstract

An image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model. Image data read from an original is acquired. A prompt for causing the proofreading server to proofread the acquired image data is generated. The acquired image data and the generated prompt are transmitted to the proofreading server, and the proofreading server performs proofreading based on the generated prompt using the learned model with respect to the acquired image data and outputs a result of the proofreading. A proofreading result output from the proofreading server is output to an output destination.

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
[0001] The present invention relates to an image processing apparatus that is capable of obtaining a result of proofreading of an original with a simple operation, a control method, and a storage medium, and more particularly to an image processing apparatus that performs proofreading by using an external server, a control method, and a storage medium.
Description of the Related Art
[0002] In recent years, Artificial Intelligence (AI) represented by Chat generative pre-trained transformer (GPT) is attracting attention. The generative AI as the recent main stream uses a learned model, referred to as the Large Language Models (LLM), and is characterized in that a text or an image is generated based on an instruction using a natural language text. As the large language model, a multimodal model, such as GPT-4.0, has also appeared which can handle not only a natural language text as an input, but also image data and voice data as an input.
[0003] The large language model learns a huge amount of data to construct a database, which requires a large capacity of storage, and hence it is general that services are built on a large-scale server.
[0004] Further, the large language model comes to be more widely used not only for generating texts and images, but also for proofreading input texts. In the case of such use form, a user is required to prepare data, which is desired to be proofread, in a terminal at hand in advance, and transmit the data from the terminal to a server in which a large language model has been constructed. In a case where data desired to be proofread is acquired from a paper medium as image data, the user is required to take time and effort for scanning an original using a scanner to form the data and taking the data into the terminal, such as a personal computer, and transmitting the data from the terminal to the server.
[0005] On the other hand, Japanese Laid-Open Patent Publication (Kokai) No. 2017-11490 discloses a technique of transmitting scanned data to a server and detecting whether proofread information exists in the image data.
[0006] However, the technique disclosed in Japanese Laid-Open Patent Publication (Kokai) No. 2017-11490 has a problem that it is necessary to manually perform proofreading of an original before scanning, and hence the burden of proofreading on a user is only partially reduced.
SUMMARY OF THE INVENTION
[0007] The present invention provides a mechanism that makes it possible to obtain a high-accuracy result of proofreading of an original of a paper medium with a simple operation.
[0008] In a first aspect of the present invention, there is provided an image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model, including an acquisition unit configured to acquire image data read from an original, a generation unit configured to generate a prompt for causing the proofreading server to proofread the acquired image data, a proofreading result request unit configured to transmit the acquired image data and the generated prompt to the proofreading server, and cause the proofreading server

to perform proofreading on the acquired image data based on the generated prompt using the learned model, and output a result of the proofreading, and an output unit configured to output a proofreading result output from the proofreading server to an output destination.
[0009] In a second aspect of the present invention, there is provided a method of controlling an image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model, including acquiring image data read from an original, generating a prompt for causing the proofreading server to proofread the acquired image data, transmitting the acquired image data and the generated prompt to the proofreading server, and causing the proofreading server to perform proofreading on the acquired image data based on the generated prompt using the learned model, and output a result of the proofreading, and outputting a proofreading result output from the proofreading server to an output destination.
[0010] According to the present invention, it is possible to obtain a high-accuracy result of proofreading of an original of a paper medium with a simple operation.
[0011] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is a diagram showing a basic configuration of a system including an image processing apparatus according to the present embodiment.
[0013] FIG. **2** is a block diagram showing a basic hardware configuration of an inside of the image processing apparatus.
[0014] FIG. **3** is a block diagram showing a basic hardware configuration of an inside of a large language model server.
[0015] FIG. **4** is a flowchart of a proofreading scan process according to the present embodiment.
[0016] FIG. **5** is a diagram showing a main menu screen displayed on a display appearing in FIG. **2**.
[0017] FIG. **6** is a diagram showing a proofreading scan screen displayed on the display.
[0018] FIG. **7** is a flowchart of a preview screen display control process.
[0019] FIG. **8** is a diagram showing a proofreading scan screen including a preview screen, which is displayed in a step in FIG. **4**.
[0020] FIG. **9** is a diagram showing an example of image data which is output as a proofreading result from the large language model server and in which proofreading items are superimposed on image data obtained by scanning an original on which a notification has been printed.
[0021] FIG. **10** is a diagram showing text data of a list of texts converted from the proofreading result shown in FIG. **9**, which is output from the large language server.
[0022] FIG. **11** is a diagram showing an example of image data which is output as a proofreading result from the large language model server and in which a proofreading item is superimposed on image data obtained by scanning an original on which a form has been printed.
[0023] FIG. **12** is a flowchart of a proofreading scan data transmission execution process performed by a central processing unit (CPU) when a proofreading scan data transmission screen is displayed.
[0024] FIG. **13** is a diagram showing a proofreading scan data transmission screen displayed in a step S**714** in FIG. **7**.
[0025] FIG. **14** is a diagram showing an AI destination setting screen.
[0026] FIG. **15** is a diagram showing a prompt generation table used in a step S**408** in FIG. **4**.
DESCRIPTION OF THE EMBODIMENTS
[0027] The present invention will now be described in detail below with reference to the

accompanying drawings showing embodiments thereof. Note that the present invention is not limited to the embodiments described below, and not all combinations of features described in the embodiments are absolutely essential to solution according to the invention.
[0028] FIG. **1** is a diagram showing a basic configuration of a system including an image processing apparatus according to the present embodiment.
[0029] Referring to FIG. **1**, one system is formed by communicably connecting the image processing apparatus, denoted by reference numeral **1**, and a large language model server **2**, via a network.
[0030] The large language model server **2** generates output data therein based on an instruction and data input, which are received from the image processing apparatus **1**, via the network, and transmits the output data to the image processing apparatus **1** via the network as a response.
[0031] FIG. **2** is a block diagram showing a basic hardware configuration of an inside of the image processing apparatus **1**.
[0032] The image processing apparatus **1** includes a CPU **201**, a graphical processing unit (GPU) **203**, an embedded MultiMediaCard (eMMC) **204**, a random access memory (RAM) **205**, a network controller **206**, a universal system bus (USB) host controller **208**, and a USB device controller **210**, as the devices connected to a system bus **202**. Further, the image processing apparatus **1** includes a display controller **212**, an input section controller **214**, a real-time clock (RTC) **216**, a serial advanced technology attachment (SATA) interface (I/F) **217**, a scanner I/F **219**, a printer I/F **221**, and a counter **223**, as the devices connected to the system bus **202**. These devices can communicate with each other via the system bus **202**.
[0033] The CPU **201** is a central processing unit that operates software for operating the image processing apparatus **1**.
[0034] The GPU **203** is a graphics processing unit that performs image processing.
[0035] The eMMC **204** stores the software (programs) of the image processing apparatus **1**, a database necessary for the operation of the image processing apparatus **1**, and a temporary storage file.
[0036] The RAM **205** is a random access memory which is used for loading a program of the image processing apparatus **1** and is used as an area for storing variables when the program operates and data transferred from each unit by direct memory access (DMA).
[0037] The network controller **206** controls a network I/F **207** for performing communication between the image processing apparatus **1** and other devices on the network. The CPU **201** transmits image data and prompts, described hereinafter, to the large language model server **2** via the network controller **206** and the network I/F **207**. Further, the CPU **201** receives a proofreading result, which is inferred by a learned model, described hereinafter, based on the image data and the prompts, and is transmitted from the large language model server **2**, via the network controller **206** and the network I/F **207**.
[0038] The USB host controller **208** controls a USB host I/F **209** for connecting to a USB device to perform communication when the image processing apparatus **1** functions as a USB host.
[0039] The USB device controller **210** controls a USB device I/F **211** for connecting to a USB host to perform communication when the image processing apparatus **1** functions as a USB device.
[0040] The display controller **212** performs display control of a display **213** that displays an operating status of the image processing apparatus **1** so as to enable a user to confirm the operating status.
[0041] The input section controller **214** controls an input section **215** that receives an instruction to the image processing apparatus **1** from a user. The input section **215** is specifically an input system including a keyboard, a mouse, a numeric key, a cursor key, a touch panel, and an operation section keyboard. In a case where the input section **215** is a touch panel, the input section **215** is physically mounted on a surface of the display **213**.
[0042] The RTC **216** is a real-time clock having a clock function, an alarm function, a timer

function, and so forth.

[0043] The serial advanced technology attachment (SATA) I/F **217** is an interface for connecting a solid state driver (SSD) **218** to the image processing apparatus **1**. The SSD **218** stores a file having a large size, temporary data of an application, and so forth.

[0044] The scanner I/F **219** is an interface for connecting a scanner **220** that reads an image from an original to the image processing apparatus **1**. The CPU **201** acquires image data of an image read from an original by using the scanner **220** from the scanner I/F **219** (acquisition unit).

[0045] The printer I/F **221** is an interface for connecting a printer **222** for printing image data to the image processing apparatus **1**.

[0046] The counter **223** records the number of copies of each print job.

[0047] FIG. **3** is a block diagram showing a basic hardware configuration of an inside of the large language model server **2**.

[0048] Referring to FIG. **3**, the large language model server **2** includes a CPU **301**, a RAM **303**, a GPU **304**, a video random access memory (VRAM) **305**, a communication section **306**, and a storage section **307**, as the devices connected to a system bus **302**. These devices can communicate with each other via the system bus **302**.

[0049] The CPU **301** operates software for operating the large language model server **2**.

[0050] The RAM **303** is used for loading a program executed by the CPU **301**, and the storage section **307** is used as an area for storing programs and data.

[0051] The communication section **306** can perform communication with external apparatuses including the image processing apparatus **1** via the network and inputs and outputs data from and to the external apparatuses.

[0052] In the storage section **307**, a model learned by machine learning (learned model) is stored. As the learned model, for example, a model using a neural network architecture, referred to as the transformer, is used.

[0053] The GPU **304** is used for a process for generating output data by performing inference using the learned model stored in the storage section **307**.

[0054] The VRAM **305** is used as a location where data to be used is loaded when the GPU **304** performs processing. When the large language model server **2** (proofreading server) receives image data and instructions, referred to as prompts, which are input from an external apparatus, via the communication section **306**, the CPU **301** uses the GPU **304** to perform inference using the learned model stored in the storage section **307**. Data output as a result of the inference (e.g. image data representing a proofreading result, described hereinafter) is transmitted to the external apparatus via the communication section **306** as a response. The large language model server **2** can receive prompts using a natural language, and this enables the external apparatus to provide flexible instructions and retrieve a variety of outputs from the learned model. Note that here, the inference is performed by using the GPU **304**, by way of example, but the CPU **301** can perform the inference instead.

[0055] FIG. **4** is a flowchart of a proofreading scan process according to the present embodiment. The present process is executed by the CPU **201** that reads out a program stored in the eMMC **204** and loads the read program into the RAM **205**.

[0056] The present process will be described below using screens shown in FIGS. **5** and **6** and displayed on the display **213**. Note that icon arrangements and words on the screens shown in FIGS. **5** and **6** are indicated by way of example and are not intended to limit the configuration of the present invention.

[0057] The present process is started from a state in which a main menu screen shown in FIG. **5** is displayed on the display **213**.

[0058] First, in a step S**401**, when a proofreading scan icon **501** on the main menu screen (see FIG. **5**) is detected to be pressed, the CPU **201** proceeds to a step S**402** and displays a proofreading scan screen (first display unit) shown in FIG. **6** on the display **213**.

[0059] As shown in FIG. **6**, on the proofreading scan screen, buttons **601** to **610** and **615** for receiving a user's instruction of proofreading contents, and a start button **611** are displayed, in a state selectable by a user.

[0060] Referring again to FIG. **4**, if at least one of the buttons **601** to **610** and **615** is detected to be pressed (YES in S**404**) before the start button **611** is pressed (NO in S**403**), the CPU **201** proceeds to a step S**405**, whereas if not (NO in S**404**), the CPU **201** returns to the step S**403**.

[0061] In the step S**405**, the CPU **201** changes the pressed state of the button detected to be pressed, from “non-pressed” to “being pressed”. Here, the “being pressed” state refers to a state in which the button in question is being selected by the user, and the “non-pressed” state refers to a state in which the button in question is not being selected by the user. The pressed state of each of the buttons **601** to **610** and **615** is stored in the RAM **205**. Note that when a button, the pressed state of which has been set to “being pressed”, is detected to be pressed again, the CPU **201** changes the pressed state of the button detected to be pressed again to “non-pressed”. Further, the “only proofreading” button **601** and the “proofreading+correction” button **602** are in an exclusive relation, and the pressed state of only one of these buttons can be changed to “being pressed”.

[0062] If the start button **611** is detected to be pressed (YES in S**403**), the CPU **201** proceeds to a step S**406** and determines whether or not one or more buttons of the buttons **603** to **610** (inspection items) are in the “being pressed” state. If no button is in the “being pressed” state (NO in S**406**), the CPU **201** displays a warning message (step S**407**), then returns to the step S**402**, and displays the proofreading scan screen again. On the other hand, if the pressed state of at least one of the buttons is “being pressed” (YES in S**406**), the CPU **201** (generation unit) generates a prompt each associated with a button in the “being pressed” state (step S**408**). Each prompt generated in this step is e.g. an instruction text using a natural language, and the CPU **201** generates a text for instructing the large language model server **2** to perform proofreading of image data.

[0063] In the present embodiment, the CPU **201** generates prompts in the step S**408** with reference to a prompt generation table (see FIG. **15**) stored in the eMMC **204** in advance.

[0064] As shown in FIG. **15**, the prompt generation table includes items **1501** to **1512** and prompts **1513** to **1524** each in a natural language, which are associated with the items **1501** to **1512**, respectively.

[0065] The items **1502** to **1509** are associated with the buttons **603** to **610**, respectively, the item **1510** is associated with the buttons **601** and **602**, the item **1511** is associated with the button **615**, and the item **1512** is associated with the button **602**.

[0066] When generating prompts in the step S**408**, the CPU **201**, first, reads a prompt **1513** corresponding to the item **1501** of “basic instruction” into the RAM **205**.

[0067] Next, in a case where any of the buttons **603** to **610** is/are in the pressed state of “being pressed”, the CPU **201** loads a prompt or prompts associated with the button(s) into the RAM **205**.

[0068] Further, in a case where the pressed state of one of the “only proofreading” button **601** and the “proofreading+correction” button **602** is “being pressed”, the CPU **201** reads a prompt **1522** of “instruction for imaging a result” into the RAM **205**. Further, in a case where the pressed state of the “proofreading+correction” button **602** (proofreading & correction instruction item) is “being pressed”, the CPU **201** reads a prompt **1524** (correction instruction prompt) of “instruction for correcting an image” into the RAM **205**. Further, in a case where the pressed state of the “conversion to text” button **615** (instruction item for conversion to a text) is “being pressed”, the CPU **201** reads a prompt **1523** (conversion-to-text instruction prompt) of “instruction for converting a result into a text” into the RAM **205**.

[0069] Finally, the CPU **201** generates one prompt by consolidating all prompts loaded into the RAM **205**.

[0070] Referring again to FIG. **4**, after generating the prompt in the step S**408**, the CPU **201** executes scanning of the original (step S**409**). The CPU **201** provides a scan instruction to the scanner **220** via the scanner I/F **210**, whereby scanning of the original is executed. Then, the CPU

201 (proofreading result request unit) stores the scanned image data in the eMMC **204** or the RAM **205**.

[0071] Next, the CPU **201** (proofreading result request unit) transmits the image data and the prompt to the large language model server **2** via the network controller **206** and the network I/F **207** (step S**410**). The data can be transmitted in a form using the Web application programming interface (API) using e.g. hypertext transfer protocol (HTTP) communication.

[0072] Upon receipt of the image data and the prompt from the image processing apparatus **1** in the step S**410**, the CPU **301** of the large language model server **2** performs proofreading of the image data based on the prompt, and outputs a result of the proofreading to the image processing apparatus **1**. This proofreading result can be such image data as shown in FIGS. **9** and **11** or such text data indicating a list of texts converted from the proofreading results as shown in FIG. **10**. The CPU **301** transmits the output proofreading result from the communication section **306** to the image processing apparatus **1**, and the CPU **201** stores the received data in the eMMC **204** or the RAM **205**. In doing this, in a case where the pressed state of the "only proofreading" button **601** is "being pressed", the CPU **201** receives only the image data which has been proofread. On the other hand, in a case where the pressed state of the "proofreading+correction" button **602** is "being pressed", the CPU **201** receives corrected image data on which an error point has been corrected based on the proofreading result together with the image data which has been proofread. In a case where the corrected image data is received together with the image data which has been proofread, the CPU **201** adds the corrected image data to the image data which has been proofread as another page. Further, in a case where the pressed state of the "conversion to text" button **615** is "being pressed", the CPU **201** receives text data generated by converting the proofreading result into a text.

[0073] In a step S**411**, the CPU **201** (output unit) stores the received proofreading result in the eMMC **204** or the RAM **205** (output destination) and displays the received proofreading result on a preview screen **801** (see FIG. **8**) on the display **213** (output destination), followed by terminating the present process. In the step S**411**, the image data of the proofreading result, shown in FIG. **9** or **11** by way of example, is displayed on the preview screen **801**.

[0074] FIG. **8** is a diagram showing the proofreading scan screen including the preview screen **801**, which is displayed in the step S**411** in FIG. **4**.

[0075] The proofreading scan screen includes the preview screen **801**, a cancel button **802**, an original data button **803**, a proofreading result button **804**, a corrected data button **805**, and page switching buttons **806** and **807**. Further, the proofreading scan screen includes a transmission button **808**, a save button **809**, and a print button **810**. These buttons are each formed by an icon selectable by the user.

[0076] Next, a flow of a process for controlling the display of the preview screen **801** displayed in the step S**411** will be described.

[0077] FIG. **7** is a flowchart of a preview screen display control process executed according to a user's operation after displaying the preview screen in the step S**411**. The present process is executed by the CPU **201** that reads out a program stored in the eMMC **204** and loads the read program into the RAM **205**.

[0078] The present process will be described below with reference to the proofreading results shown in FIGS. **9** to **11**, each of which is received from the large language model server **2** and is displayed on the preview screen **801** by the CPU **201**. Note that image data shown in each of FIGS. **9** to **11** is only example and is not intended to limit the scope of the present invention.

[0079] First, in a step S**701**, the CPU **201** detects whether or not a user's operation has been performed on the proofreading scan screen. If a user's operation has been performed (YES in S**701**), the CPU **201** proceeds to a step S**702**. On the other hand, if no user's operation has been performed (NO in S**701**), the CPU **201** waits until a user's operation is performed.

[0080] If the cancel button **802** is detected to be pressed (YES in S**702**), the CPU **201** changes the preview screen **801** of the proofreading scan screen, shown in FIG. **8**, to a blank space display (step S**703**), followed by terminating the present process. On the other hand, if the cancel button **802** has not been pressed (NO in S**702**), the CPU **201** proceeds to a step S**704** and detects whether or not the original data button **803** has been pressed.

[0081] If the original data button **803** is detected to be pressed (YES in S**704**), the CPU **201** displays the data of the original, which has been scanned in the step S**409** but has not been proofread, on the preview screen **801** (step S**705**), and then proceeds to a step S**706**. On the other hand, if the original data button **803** has not been pressed (NO in S**704**), the CPU **201** directly proceeds to the step S**706** and detects whether or not the proofreading result button **804** has been pressed.

[0082] If the proofreading result button **804** is detected to be pressed (YES in S**706**), the CPU **201** displays the proofreading result output from the large language model server **2** on the preview screen **801** (step S**707**), and then proceeds to a step S**708**. On the other hand, if the proofreading result button **804** has not been pressed (NO in S**706**), the CPU **201** directly proceeds to the step S**708** and detects whether or not the corrected data button **805** has been pressed.

[0083] FIG. **9** is a diagram showing an example of image data **900** output from the large language model server **2** as a proofreading result, in which proofreading items are superimposed on image data obtained by scanning an original on which a notification has been printed. The image data **900** is displayed on the preview screen **801** by the CPU **201** when the proofreading result button **804** (see FIG. **8**) is pressed, and proofreading items are associated with respective pointed parts each surrounded by a frame, as illustrated by proofreading items **901** to **904**. Note that the pointed part can be indicated by using a method other than the method of using a frame surrounding a pointed part as shown in FIG. **9** insofar as the pointed part is highlighted in the image data **900**. For example, the pointed part can be displayed by bold characters or shown in a flashing display.

[0084] FIG. **11** is a diagram showing an example of image data **1100** output from the large language model server **2** as a proofreading result, in which the proofreading item is superimposed on image data obtained by scanning an original on which a form has been printed. Similar to the image data **900** shown in FIG. **9**, the image data **1100** shown in FIG. **11** is displayed on the preview screen **801** when the proofreading result button **804** (see FIG. **8**) is pressed, and a proofreading item is associated with a pointed part (error in value in this example) surrounded by a frame, as illustrated by a proofreading item **1101**.

[0085] Referring again to FIG. **7**, if the corrected data button **805** is detected to be pressed (YES in S**708**), the CPU **201** detects whether or not corrected image data exists in the eMMC **204** or the RAM **205** (step S**709**). If the corrected image data exists (YES in S**709**), the CPU **201** displays the corrected image data (step S**710**) and proceeds to a step S**711**. On the other hand, if the corrected data button **805** has not been pressed or the corrected data does not exist (NO in S**708** or NO in S**709**), the CPU **201** directly proceeds to the step S**711**.

[0086] In the step S**711**, the CPU **201** detects whether or not one of the page switching buttons **806** and **807** has been pressed.

[0087] The CPU **201** detects whether or not one of the page switching buttons **806** and **807** has been pressed (a user's instruction has been provided) (step S**711**). If one of the page switching buttons **806** and **807** has been pressed (YES in S**711**), the CPU **201** changes the image data being displayed on the preview screen **801** to display another page (such as the corrected image data or image data **1000** (see FIG. **10**)) of the image data (step S**712**). On the other hand, if none of the page switching buttons **806** and **807** has been pressed (NO in S**711**), the CPU **201** proceeds to a step **713** and detects whether or not the transmission button **808** has been pressed.

[0088] FIG. **10** is a diagram showing the image data **1000** which is output from the large language model server **2** and illustrates a list of texts converted from the proofreading result shown in FIG. **9**. When the proofreading result shown in FIG. **9** is received from the large language model server **2**, the CPU **201** (conversion unit) converts the proofreading result to the image data **1000**. Then, the

CPU **201** adds the image data **1000** to the proofreading result (image data **900**) output from the large language model server **2** as another page. Further, if one of the page switching buttons **806** and **807** (see FIG. **8**) is pressed when the image data **900** shown in FIG. **9** is being displayed on the preview screen **801**, the CPU **201** changes the preview screen **801** to the image data **1000** shown in FIG. **10**. In the image data **1000**, the pointed items each formed by a pointed part and a pointed content in the image data **900** are displayed in a list, as illustrated by pointed items **1001** to **1004**.

[0089] Referring again to FIG. **7**, if the transmission button **808** is detected to be pressed (YES in S**713**), the CPU **201** proceeds to a step S**714**. In the step S**714**, the CPU **201** displays a proofreading scan data transmission screen **1300** (see FIG. **13**) for displaying a plurality of destinations which can each be set as the transmission destination in a state selectable by a user, followed by terminating the present process. The user can transmit the original data and the image data **900** (the corrected data can also be transmitted, if acquired) to a desired destination by operating the proofreading scan data transmission screen **1300**. On the other hand, if the transmission button **808** has not been pressed (NO in S**713**), the CPU **201** proceeds to a step S**715** and detects whether or not the save button **809** has been pressed. Note that the proofreading scan data is one of data of an original which has been scanned but not proofread, image data output from the large language model server **2** as a proofreading result, and image data on which an error point has been corrected based on the proofreading result output from the large language model server **2**.

[0090] If it the save button **809** is detected to be pressed (YES in S**715**), the CPU **201** stores the scanned image data and the image data output from the large language model server **2** in a predetermined area of the eMMC **204** (step S**716**), followed by terminating the present process. On the other hand, if the save button **809** has not been pressed (NO in S**715**), the CPU **201** proceeds to a step S**717** and detects whether or not the print button **810** has been pressed.

[0091] If the print button **810** is detected to be pressed (YES in S**717**), the CPU **201** proceeds to a step S**718**. On the other hand, if the print button **810** has not been pressed (NO in S**717**), the CPU **201** returns to the step S**702**.

[0092] In the step S**718**, the CPU **201** transmits the image data to the printer **222** via the printer I/F **221** to execute printing, followed by terminating the present process. Note that the image data to be printed can be designated by the user.

[0093] Next, a flow of a proofreading scan data transmission execution process executed when the proofreading scan data transmission screen **1300** is displayed in the step S**714** will be described.

[0094] FIG. **12** is the flowchart of the proofreading scan data transmission execution process performed by the CPU **201** when the proofreading scan data transmission screen **1300** is being displayed. The present process is executed by the CPU **201** that reads out a program stored in the eMMC **204** and loads the read program into the RAM **205**.

[0095] First, in a step S**1200**, the CPU **201** detects whether or not a user's operation has been performed on the proofreading scan data transmission screen **1300**. If a user's operation has been performed (YES in S**1200**), the CPU **201** proceeds to a step S**1201** and detects whether or not a return button **1310** has been pressed. On the other hand, if no user's operation has been performed (NO in S**1200**), the CPU **201** waits until a user's operation is performed.

[0096] If the return button **1310** is detected to be pressed (YES in S**1201**), the CPU **201** displays the preview screen **801** (step S**1202**), followed by terminating the present process. On the other hand, if the return button **1310** has not been pressed (NO in S**1201**), the CPU **201** proceeds to a step S**1203** and detects whether or not an address book button **1302** has been pressed.

[0097] If the address book button **1302** is detected to be pressed (YES in S**1203**), the CPU **201** receives an address set from the address book to set the transmission destination to the received address (step S**1204**), and the CPU **201** proceeds to a step S**1205**. On the other hand, if the address book button **1302** has not been pressed (NO in S**1203**), the CPU **201** directly proceeds to the step S**1205** and detects whether or not a to-myself button **1303** has been pressed.

[0098] If it the to-myself button **1303** is detected to be pressed (YES in S**1205**), the CPU **201** sets a mail address of the user who is operating the image processing apparatus **1** (step S**1206**) and proceeds to a step S**1207**. On the other hand, if the to-myself button **1303** has not been pressed (NO in S**1205**), the CPU **201** directly proceeds to the step S**1207** and detects whether or not a to-cloud button **1304** has been pressed.

[0099] If the to-cloud button **1304** is detected to be pressed (YES in S**1207**), the CPU **201** receives the URL of the cloud and sets the destination to the received URL (step S**1208**), whereafter the CPU **201** proceeds to a step S**1209**. On the other hand, if the to-cloud button **1304** has not been pressed (NO in S**1207**), the CPU **201** directly proceeds to the step S**1209** and detects whether or not a fax button **1305** has been pressed.

[0100] If the fax button **1305** is detected to be pressed (YES in S**1209**), the CPU **201** receives the destination of the fax and sets the transmission destination to the received destination (step S**1210**), whereafter the CPU **201** proceeds to a step S**1211**. On the other hand, if the fax button **1305** has not been pressed (NO in S**1209**), the CPU **201** directly proceeds to the step S**1211** and detects whether or not an original data button **1307** has been pressed.

[0101] If the original data button **1307** is detected to be pressed (YES in S**1211**), the CPU **201** performs setting so as to use the image data (original data) which has not been changed after being scanned, as the transmission data (step S**1212**), and then proceeds to a step S**1213**. On the other hand, if the original data button **1307** has not been pressed (NO in S**1211**), the CPU **201** directly proceeds to the step S**1213** and detects whether or not a proofreading result button **1308** has been pressed.

[0102] If the proofreading result button **1308** is detected to be pressed (YES in S**1213**), the CPU **201** performs setting to use the proofreading result data (the image data shown in FIG. **9** or **10** in this example) as the transmission data (step S**1214**) and proceeds to a step S**1215**. On the other hand, if the proofreading result button **1308** has not been pressed (NO in S**1213**), the CPU **201** directly proceeds to the step S**1215** and detects whether or not a corrected data button **1309** has been pressed. If the corrected data button **1309** has not been pressed (NO in S**1215**), the CPU **201** proceeds to a step S**1218** and detects whether or not an execution button **1306** has been pressed.

[0103] On the other hand, if the corrected data button **1309** is detected to be pressed (YES in S**1215**), the CPU **201** detects whether or not the corrected data exists (step S**1216**). If the corrected data does not exist (NO in S**1216**), the CPU **201** displays a warning message (step S**1219**) and then proceeds to the step S**1218**. On the other hand, if the corrected data exists (YES in S**1216**), the CPU **201** performs setting so as to use the corrected data as the transmission data (step S**1217**) and then proceeds to the step S**1218**.

[0104] If the execution button **1306** is detected to be pressed (YES in S**1218**), the CPU **201** executes transmission of the transmission data set as the transmission data to the transmission destination set as the destination (step S**1220**), followed by terminating the present process. On the other hand, if the execution button **1306** has not been pressed (NO in S**1218**), the CPU **201** returns to the step S**1200**.

[0105] FIG. **14** is a diagram showing an AI destination setting screen.

[0106] Here, the AI destination setting screen (destination setting user interface (UI)) is a screen on which a setting of access to the large language model server **2** used for proofreading can be set. This screen is displayed in a case where an AI destination setting menu is selected on a menu screen (not shown) displayed when a set button **1311** on the proofreading scan data transmission screen **1300** (see FIG. **13**) is pressed.

[0107] Referring to FIG. **14**, the AI destination setting screen includes an address setting section **1401**, an ID setting section **1402**, a password setting section **1403**, and a save button **1404**.

[0108] In the address setting section **1401**, an internet protocol (IP) address, a URL, or the like is set. In the ID setting section **1402** and the password setting section **1403**, an ID and a password, which are required for authentication, are set, respectively, when a service of the large language model server **2** is used. When accessing the large language model server **2** set by the address setting section **1401**, the CPU **201** performs authentication by using the information input to the ID setting section **1402** and the password setting section **1403**.

[0109] As described above, in the present embodiment, the proofreading scan screen is displayed in the step S**402** in FIG. **4**, the setting of the inspection item is received in the steps S**404** and S**405**, and the scanned image is transmitted to the large language model server **2** to execute proofreading in the steps $408 to S**411**. With this, the user can obtain a high-accuracy result of proofreading with a simple operation.

[0110] Further, by setting the address of the large language model server **2**, and the ID and the password for authentication, which are required for accessing the large language model server **2**, in advance, it is possible to smoothly perform the processing in the step S**410**.

OTHER EMBODIMENTS

[0111] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0112] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0113] This application claims the benefit of Japanese Patent Application No. 2024-035859 filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model, comprising: an acquisition unit configured to acquire image data read from an original; a generation unit configured to generate a prompt for causing the proofreading server to proofread the acquired image data; a proofreading result request unit configured to transmit the acquired image data and the generated prompt to the proofreading server, and cause the proofreading server to perform proofreading on the acquired image data based on the generated prompt using the learned model, and output a result of the proofreading; and an output unit configured to output a proofreading result output from the proofreading server to an output destination.

2. The image processing apparatus according to claim 1, wherein the proofreading server is a large language model server.

3. The image processing apparatus according to claim 1, wherein the output unit displays image

data representing the proofreading result on a preview screen.
4. The image processing apparatus according to claim 1, wherein the output unit transmits image data representing the proofreading result to a predetermined destination.
5. The image processing apparatus according to claim 1, wherein the generated prompt is a text using a natural language.
6. The image processing apparatus according to claim 1, further comprising a first display unit configured to display a plurality of inspection items for generating the prompt in a state selectable by a user, and wherein the generation unit generates the prompt according to an inspection item or inspection items selected by a user from the plurality of inspection items.
7. The image processing apparatus according to claim 6, wherein the first display unit further displays a proofreading & correction instruction item for instructing proofreading and correction in a state selectable by a user, and wherein in a case where the proofreading & correction instruction item is selected by a user, the generation unit further generates a correction instruction prompt for instructing correction of the acquired image data, and the proofreading result request unit further transmits the correction instruction prompt to the proofreading server and causes the proofreading server to generate image data on which an error part has been corrected based on the proofreading result, and output the corrected image data.
8. The image processing apparatus according to claim 7, wherein the output unit displays one of the corrected image data and the image data representing the proofreading result after switching therebetween, on a preview screen, according to a user's instruction.
9. The image processing apparatus according to claim 6, wherein the first display unit further displays a text conversion instruction item for instructing acquisition of a list of texts converted from the proofreading result in a state selectable by a user, and wherein in a case where the text conversion instruction item is selected by a user, the generation unit further generates a conversion-to-text instruction prompt for instructing acquisition of a list of texts converted from the proofreading result, and the proofreading result request unit further transmits the conversion-to-text instruction prompt to the proofreading server and causes the proofreading server to generate a list of texts converted from the proofreading result and output the list of the texts.
10. The image processing apparatus according to claim 9, further comprising a conversion unit configured to convert the list of the texts to image data, and wherein the output unit displays one of the converted image data and the image data representing the proofreading result after switching therebetween, on a preview screen, according to a user's instruction.
11. The image processing apparatus according to claim 1, further comprising a destination setting UI for making a setting of access to the proofreading server, and wherein the proofreading result request unit performs communication with the proofreading server by using the setting of access to the proofreading server, which has been made by the destination setting UI.
12. The image processing apparatus according to claim 3, wherein in the image data representing the proofreading result, a part pointed by the proofreading is highlighted and a proofreading item is displayed in a state associated with the pointed part.
13. The image processing apparatus according to claim 1, further comprising a second display unit configured to further display a plurality of destinations which can be set as the predetermined destination in a state selectable by a user.
14. A method of controlling an image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model, comprising: acquiring image data read from an original; generating a prompt for causing the proofreading server to proofread the acquired image data; transmitting the acquired image data and the generated prompt to the proofreading server, and causing the proofreading server to perform proofreading on the acquired image data based on the generated prompt using the learned model, and output a result of the proofreading; and outputting a proofreading result output from the proofreading server to an output destination.

15. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a method of controlling an image processing apparatus that communicably connects to a proofreading server that performs inference using a learned model, wherein the method comprises: acquiring image data read from an original; generating a prompt for causing the proofreading server to proofread the acquired image data; transmitting the acquired image data and the generated prompt to the proofreading server, and causing the proofreading server to perform proofreading on the acquired image data based on the generated prompt using the learned model, and output a result of the proofreading; and outputting a proofreading result output from the proofreading server to an output destination.